MIL-S-3419B
21 February 1968
SUPERSEDING
MIL-P-3419A (CE)
15 September 1960

MILITARY SPECIFICATION

SHOP EQUIPMENT,

CARTOGRAPHIC INSTRUMENT REPAIR,

FIELD MAINTENANCE, PACKAGING OF

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the requirements for materials, methods, and procedures for the preservation, packaging, and packing of SHOP EQUIPMENT, CARTOGRAPHIC INSTRUMENT REPAIR, FIELD MAINTENANCE SC6675-93-CL-E12. (FSN 6675-641-3623 or FSN 6675-926-1243 for MAP use).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-621	- Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	- Box, Fiberboard.
PPP-C-650	- Crates, Wood, Open and Covered.
PPP-F-320	- Fiberboard; Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes.
PPP-T-97	- Tape, Pressure-Sensitive Adhesive, Filament Reinforced.

FSC 6675

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Military

MIL-C-104	- Crates, Wood; Lumber and Plywood Sheathed, Nailed and Bolted.
MIL-P-116	- Preservation, Methods of.
MIL-H-15424	- Hand Tools; Packaging of.
MIL-E-16298	- Electric Machines having Rotating Parts and Associated Repair Parts: Packaging of.
MIL-M-18058	- Preparation for Delivery of Machine Tools (Metal and Woodworking), Accessories and Repair Parts.

STANDARDS

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-1186	- Cushioning, Anchoring, Bracing, Blocking and Waterproofing; with Appropriate Test Methods.

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

UNIFORM CLASSIFICATION COMMITTEE

Uniform Freight Classification Rules.

(Application for copies should be addressed to the Uniform Classification Committee, 202 Union Station, 516 West Jackson boulevard, Chicago, Illinois 60606.)

AMERICAN TRUCKING ASSOCIATIONS, INC.

National Motor Freight Classification Rules.

(Application for copies should be addressed to the American Trucking Associations, Inc., 1616 P Street NW, Washington, D. C. 20036.)

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3. REQUIREMENTS

3.1 Materials. Materials shall be as specified herein and in the applicable specifications. Materials not specified shall be selected by the supplier and shall be subject to all provisions of this specification.

3.2 Preproduction pack. The supplier shall furnish a preproduction pack for examination and test within the time frame specified (see 6.2) to prove, prior to starting production packaging and packing, that the applied preservation, packaging, packing, and marking comply with the requirements of this specification (see 6.3). Examinations and tests shall be those specified herein.

3.3 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.2).

3.3.1 Level A.

3.3.1.1 Preservatives. Preservatives specified herein shall conform to the applicable specification referenced in and shall be applied in accordance with MIL-P-116.

3.3.1.2 Components.

3.3.1.2.1 Hand tools and tool sets. Hand tools and tool sets shall be preserved and packaged in accordance with the level A requirements of MIL-H-15424. Hand tools and tool sets not specifically identified in MIL-H-15424 shall be preserved and packaged as specified for similar items.

3.3.1.2.2 Machine tools. Machine tools with accessories, repair parts, operating tools, and technical publications shall be preserved and packaged in accordance with the level A requirements of MIL-M-18058.

3.3.1.2.3 Desk light. The desk light shall be preserved in accordance with MIL-P-116, method III, using a close-fitting fiberboard box conforming to PPP-B-636, W6c, style optional. The box shall be closed and waterproof sealed with tape in accordance with the appendix to the box specification.

3.3.1.2.4 Goggles. The goggles, in vendor carton when furnished, shall be preserved in accordance with MIL-P-116, method IC1.

3.3.1.2.5 Lubricating oil. The bottles of lubricating oil shall be preserved together in accordance with MIL-P-116, method III, using a close-fitting box conforming to PPP-B-636, W6c, style optional. The bottles shall be separated and cushioned with pads or corrugated fiberboard conforming to PPP-F-320, type CF, variety SW, grade W6c. Lubricating oil furnished in cans shall be packaged in a box conforming to PPP-B-636, W6c, style optional.

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3.3.1.2.6 Sharpening stones. Each stone shall be preserved in accordance with MIL-P-116, method IC1.

3.3.1.2.7 Steel bars and brass rods. Steel bars shall be coated with type P-1 preservative. The steel bars shall be bundled together and the brass rods shall be bundled together; each bundle shall be secured with equally spaced wraps of tape conforming to PPP-T-97, type IV.

3.3.1.2.8 Generator set. The generator set shall be preserved in accordance with MIL-E-16298, level A, method IIa. The generator set shall be secured directly through the barrier material to the base of a box conforming to PPP-B-601, overseas type, style optional. The box shall be closed in accordance with the appendix to the box specification.

3.3.1.2.9 Calibration kits. Each calibration kit shall be preserved in accordance with MIL-P-116, method IIb. The carrying case shall be the inner container, and the outer container shall conform to PPP-B-636, W5c, style optional. The box shall be closed and waterproof sealed with tape in accordance with the appendix to the box specification.

3.3.1.3 Intermediate packaging.

3.3.1.3.1 Machine tools. Each complete lathe, drilling machine, and grinder shall be intermediate packaged in a close-fitting box conforming to PPP-B-621, class 1, style optional. Blocking and anchoring shall be provided to secure the machine tools in place within the box.

3.3.1.3.2 Other component items. All other component items except the steel bars and brass rods shall be intermediate packaged in close-fitting boxes conforming to PPP-B-601, domestic type, style optional or PPP-B-621, class 1, style optional. The gross weight of each box shall not exceed 200 pounds.

3.3.2 Level C. The components and machine tools of the shop equipment set shall be preserved and packaged to afford protection against deterioration and damage during shipment from the supplier to the initial destination for immediate use.

3.4 Packing. Packing shall be level A, B, or C as specified (see 6.2).

3.4.1 Level A. The components and machine tools of each complete set, packaged as specified in 3.3, shall be packed in a crate conforming to MIL-C-104, type I, class 1 or 2, style a. Blocking, bracing, and anchoring shall be in accordance with the appendix to the crate specification and in accordance with MIL-STD-1186. Strapping shall be zinc coated.

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3.4.2 Level B. The components and machine tools of each complete set, packaged as specified in 3.3 shall be packed in a crate conforming to PPP-C-650, type V, style B. Blocking, bracing, anchoring, and waterproofing shall be in accordance with the appendix to the crate specification and MIL-STD-1186. Strapping need not be zinc coated.

3.4.3 Level C. Each complete set shall be packed to assure carrier acceptance and safe delivery to destination at lowest rates in containers complying with Uniform Freight Classification Rules or National Motor Freight Classification Rules.

3.5 Marking. Marking shall be in accordance with MIL-STD-129.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Material inspection. The supplier is responsible for insuring that supplies and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards.

4.2 Classification of inspection. Inspection shall be classified as follows:

(a) Preproduction inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 Preproduction inspection.

4.3.1 Examination. The pack shall be examined for defects specified in 4.4.3. Presence of one or more defects shall be cause for rejection.

4.3.2 Test. The pack shall be subjected to any one of the guided-impact tests specified in appendix A to MIL-STD-1186. For the guided-impact test (railroad car), the car shall strike a string of five empty cars with draft gear extended and brakes set. The speed of the car at impact shall be not less than 10 miles per hour or not more than 11 miles per hour. Shifting of contents, visible damage to the contents, loosening or breaking of anchoring, blocking, or bracing within the pack shall constitute failure of the test.

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4.4 Quality conformance inspection.

4.4.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.4.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.4.3 Examination. Samples selected in accordance with 4.4.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

101.	Materials, methods, and containers not as specified. Each incorrect material, method, or container shall constitute one defect.
102.	Containers not waterproof sealed as specified.
103.	Bottles not separated and cushioned with fiberboard as specified.
104.	Rods not coated and taped as specified.
105.	Intermediate packaging not as specified.
106.	Strapping not zinc coated for level A.
107.	Marking illegible, incomplete, incorrect, or missing.

5. PREPARATION FOR DELIVERY

(Not applicable to this specification.)

6. NOTES

6.1 Intended use. This specification is intended to be used for direct reference in contracts or orders for the preservation, packaging, packing, and marking of a shop equipment set for cartographic instrument repair conforming to SC 6675-93-CL-E12. (FSN 6675-641-3623 or 6675-926-1243 for MAP use).

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Time frame required for submission of the preproduction pack (see 3.2).
(c) Level of preservation and packaging and level of packing required (see 3.3 and 3.4).

6.3 Preproduction pack. Any changes or deviations of production packs from the approved preproduction pack will be subject to the approval of the contracting officer. Approval of the preproduction pack will not relieve the supplier of his obligation to preserve, package, pack, and mark the shop equipment set in accordance with this specification.

Custodians:	Review activities:	Preparing activity:
Army - ME	Army - SM	Army - ME
Navy - YD	Navy - YD	

Project No. 6675-0296

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DEPARTMENT OF THE ARMY
U. A. ARMY MOBILITY EQUIPMENT COMMAND
RESEARCH, DEVELOPMENT AND ENGINEERING DIRECTORATE
ATTN: SMEFB-RDE-KX
FORT BELVOIR, VIRGINIA 22060

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION MIL-S-3419B Shop Equipment, Cartographic Instrument Repair, Field Maintenance, Packaging Of

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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